

Activity 2

① Stack and Queue Operations

Stack operation (LIFO)

push(54) $\rightarrow$ [54]push(52) $\rightarrow$ [54, 52]pop() $\rightarrow$ [54]push(55) $\rightarrow$ [54, 55]push(62) $\rightarrow$ [54, 55, 62]s = pop() $\rightarrow$ s = 62

Stack = [54, 55]

Queue operations (FIFO)

enqueue(21) $\rightarrow$ [21]enqueue(24) $\rightarrow$ [21, 24]dequeue() $\rightarrow$ [24]enqueue(28) $\rightarrow$ [24, 28]enqueue(32) $\rightarrow$ [24, 28, 32]

q = dequeue()

q = 24

Queue [28, 32]

Value $s + q = 62 + 24 = 86$ ② Expression $105 + 606 / * 8 - 8 = 142$

Expression	Stack	Output
10, 5		
+	+	$10 + 5 = 15$
60, 6		
/	/	$60 / 6 = 10$
*	*	$15 \times 10 = 150$
- 8	-	$150 - 8 = 142$

③ Minimum of queue is 0
Elements can be moved from Q_1 to Q_2 by repeatedly dequeuing from Q_1 and enqueueing to Q_2

④ Modified stack with reverse can be implemented using a queue and can implement a FIFO queue with $O(1)$ time per operation

⑤ Expression $8, 2, 3, ^, /, 2, 3, +, *, 5, 1, ^, -$

Push 8 $\rightarrow [8]$

Push 2 $\rightarrow [8, 2]$

Push 3 $\rightarrow [8, 2, 3]$

pop 3 and 2 $\rightarrow [8, 8]$ $(^)$ $\rightarrow 2^3 = 8$

pop 8 and 8 $\rightarrow [1]$ $(/)$ $8/8 = 1$

Push 2 $\rightarrow [1, 2]$

Push 3 $\rightarrow [1, 2, 3]$

pop 3 and 2 $\rightarrow [1, 6]$ $2 \times 3 = 6$

Top element: 6 $\Rightarrow [6, 1]$

Second element: 1

6. $a + b \times c - d \wedge e \wedge f$

Precedence Rules

Highest $\wedge$

Next $*$

lowest $+$ and $-$

$((a + b \times c) - (d \wedge (e \wedge f)))$

$((a + (b \times c)) - (d \wedge (e f \wedge)))$

$((a + (b \times c)) - d e f \wedge \wedge)$

$(a + b c \wedge - d e f \wedge \wedge)$

$\Rightarrow a b c \wedge + d e f \wedge \wedge -$

7) The sequence of values popped out

operation	stack	popped value
push(10)	[10]	
push(20)	[10, 20]	
pop	[10]	20
push(10)	[10, 10]	
push(20)	[10, 10, 20]	
pop	[10, 10]	20
pop	[10]	10
pop	[]	10
push(20)	[20]	
pop	[]	20

Sequence: 20, 20, 10, 10, 20